

Bivariate Probability Distribution:

The probability distribution of two-dimensional random variable (X, Y) is called bivariate distribution. That is, the distribution of joint probabilities $P(X=x_i, Y=y_j)$ of two random variables where $i=1, 2, \dots, m$, $j=1, 2, \dots, n$ is known as bivariate probability distribution.

The bivariate distribution of discrete random variable (X, Y) is the bivariate distribution of discrete type and the bivariate distribution of continuous random variable (X, Y) is called the bivariate continuous distribution.

Joint prob. mass function (Joint pmf):

Let (X, Y) be a two dimensional discrete random variable taking pair values

(x_i, y_j) , $i=1, 2, \dots, m$ and $j=1, 2, \dots, n$

then the probability function $P(X=x_i, Y=y_j)$ is called joint pmf of random variable (X, Y) if it satisfies the following conditions.

(i) $P(X=x_i, Y=y_j) \geq 0$

(ii) $\sum_{i=1}^m \sum_{j=1}^n P(X=x_i, Y=y_j) = 1$

The joint pmf of (X, Y) is,

$X \backslash Y$	y_1	y_2	$\dots$	y_j	$\dots$	y_n	Total
x_1	P_{11}	P_{12}		P_{1j}		P_{1n}	$P_{1.}$
x_2	P_{21}	P_{22}		P_{2j}		P_{2n}	$P_{2.}$
$\vdots$							
x_i							
$\vdots$							
x_m	P_{m1}	P_{m2}		P_{mj}		P_{mn}	$P_{m.}$
Total	$P_{.1}$	$P_{.2}$	$\vdots$	$P_{.j}$	$\vdots$	$P_{.n}$	1

Example: For the following Bivariate prob. distⁿ of X and Y

X \ Y	0	1	2	3	Total / Marginal of X
0	1/42	1/21	1/14		$p(X=0) = 1/7$
1	1/21	1/14	1/21	5/42	$p(X=1) = 2/7$
2	2/21	5/42	1/7	1/6	$p(X=2) = 10/21$
3					

Marginal / Total
 $p(Y=0) = 5/21$
 $p(Y=1) = 5/21$
 $p(Y=2) = 5/21$
 $p(Y=3) = 5/21$

Find (i) $p(X \leq 1, Y = 1)$

(ii) $p(X < 2, Y \leq 2)$

(iii) $p(X \leq 1)$

(iv) $p(Y = 2)$

(v) $p(X \geq 1, Y \leq 2)$

Sol:

$$(i) p(X \leq 1, Y = 1) = p(X=0, Y=1) + p(X=1, Y=1) \\ = \frac{1}{42} + \frac{1}{14} = \frac{2}{21}$$

$$(ii) p(X \leq 2, Y \leq 2) = p(X=0, Y \leq 2) + p(X=1, Y \leq 2) \\ = p(X=0, Y=0) + p(X=0, Y=1) + p(X=0, Y=2) \\ + p(X=1, Y=0) + p(X=1, Y=1) + p(X=1, Y=2) \\ = \frac{1}{42} + \frac{1}{21} + \frac{2}{21} + \frac{1}{21} + \frac{1}{14} + \frac{1}{21} = \frac{10}{21}$$

$$(iii) p(X \leq 1) = p(X=0) + p(X=1) = \frac{1}{7} + \frac{2}{7} = \frac{3}{7}$$

$$(iv) p(Y = 2) = \frac{10}{21}$$

$$(v) p(X \geq 1, Y \leq 2) = p(X=1, Y \leq 2) + p(X=2, Y \leq 2) \\ = p(X=1, Y=0) + p(X=1, Y=1) \\ + p(X=1, Y=2) + p(X=2, Y=0) + p(X=2, Y=1) \\ + p(X=2, Y=2) = \frac{1}{21} + \frac{1}{14} + \frac{1}{21} + \frac{2}{21} + \frac{1}{14} + \frac{1}{21} = \frac{10}{21}$$

Joint Probability density function (Joint pdf):

Let (X, Y) be a two dimensional continuous random variable then a continuous function $f(x, y)$ is said to be joint pdf of random variable (X, Y) if it satisfies the following conditions.

(i) $f(x, y) \geq 0$ for all x and y

(ii) $\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y) dx \right] dy = 1$

Example: Let (X, Y) be a two continuous random variable with joint pdf

$$f(x, y) = k(2x + 3y), \quad 0 \leq x \leq 1 \\ 0 \leq y \leq 1$$

$$= 0, \text{ otherwise}$$

Find:

(i) value of k

(ii) $P(X < 0.4, Y < 0.4) = 0.064$

(iii) $P(X > 0.5, Y < 0.5) = 0.102$

(i) we have,

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y) dx \right] dy = 1$$

$$\int_0^1 \left[\int_0^1 k(2x + 3y) dx \right] dy = 1$$

$$\int_0^1 \left[2k \int_0^1 x dx + 3ky \int_0^1 dx \right] dy = 1$$

$$\int_0^1 \left[2k \left(\frac{x^2}{2} \right)_0^1 + 3ky (x)_0^1 \right] dy = 1$$

$$\int_0^1 (k + 3ky) dy = 1$$

$$k \int_0^1 dy + 3k \int_0^1 y dy = 1$$

$$K(y)_0^1 + 3K\left(\frac{y^2}{2}\right)_0^1 = 1$$

$$Ky + \frac{3K}{2} = 1$$

$$2K + 3K = 2$$

$$5K = 2$$

$$\Rightarrow \boxed{K = 2/5}$$

(ii) $P(x < 0.4, y < 0.4)$

we have,

$$= \int_0^{0.4} \left(\int_0^{0.4} f(x,y) dx \right) dy$$

$$= \int_0^{0.4} \left[\int_0^{0.4} \frac{2}{5}(2x+3y) dx \right] dy$$

$$= \int_0^{0.4} \left[\frac{4}{5} \int_0^{0.4} x dx + \frac{6y}{5} \int_0^{0.4} dx \right] dy$$

$$= \int_0^{0.4} \left[\frac{4}{5} \left[\frac{x^2}{2} \right]_0^{0.4} + \frac{6y}{5} [x]_0^{0.4} \right] dy$$

$$= \int_0^{0.4} \left[0.064 + \frac{6}{5} y \times 0.4 \right] dy$$

$$= \int_0^{0.4} [0.064 + 0.48y] dy$$

$$= \overset{0.064}{0.016} \times 0.4 + 0.48 \times \frac{0.4^2}{2}$$

$$= \overset{0.064}{0.016} \times 0.16 + 0.0384$$

$$= \overset{0.064}{0.016} \times 0.064$$

① $P(X > 0.8, Y < 0.5)$

we have,

$$\int_{0.8}^1 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$$

$$= \int_0^{0.5} \left[\int_{0.8}^1 \frac{2}{5} (2x + 3y) dx \right] dy$$

$$= \int_0^{0.5} \left[\frac{4}{5} \int_{0.8}^1 x dx + \frac{6}{5} y \int_{0.8}^1 dx \right] dy$$

$$= \int_0^{0.5} \left[\frac{4}{5} \left[\frac{x^2}{2} \right]_{0.8}^1 + \frac{6}{5} y [x]_{0.8}^1 \right] dy$$

$$= \int_0^{0.5} \left[\frac{4}{5} \left(\frac{1}{2} - \frac{0.8^2}{2} \right) + \frac{6}{5} y (1 - 0.8) \right] dy$$

$$= \int_0^{0.5} [0.144 + 0.24y] dy$$

$$= 0.144 \int_0^{0.5} dy + 0.24 \int_0^{0.5} y dy$$

$$= 0.144 \left[y \right]_0^{0.5} + 0.24 \left[\frac{y^2}{2} \right]_0^{0.5}$$

$$= 0.144 \times 0.5 + 0.24 \times \frac{0.5^2}{2}$$

$$= 0.072 + 0.03$$

$$= 0.102$$

Joint Probability Distribution Function of X and Y :

Let (X, Y) be a two dimensional random variable and then the joint probability distribution function of X and Y is defined as the probability of the event $P(X \leq x, Y \leq y)$ for all (x, y) in XY -plane.

Therefore, the joint probability distribution function or joint cumulative distribution function (joint cdf) is given by,

$$F(x, y) = P(X \leq x, Y \leq y) \\ = \sum_{x=-\infty}^x \sum_{y=-\infty}^y P(X=x, Y=y)$$

if X and Y is discrete random variable,

$$= \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx$$

if (X, Y) is continuous random variable

Conditional Probability Mass Function of X and Y :

Let (X, Y) be a two dimensional random variable of discrete type with joint pmf $P(X=x, Y=y)$ then the conditional probability mass function of X for given $Y=y$ is defined as,

$$P(X=x | Y=y) = \frac{P(X=x \cap Y=y)}{P(Y=y)}$$

$P(Y=y)$

←
marginal

Similarly,

the conditional discrete probability function of Y for given $X = x_i$ is defined as,

$$P(Y = y_j | X = x_i) = \frac{P(X = x_i \cap Y = y_j)}{P(X = x_i)}$$

$\Rightarrow$ can also be written as,

$$= \frac{P(X = x_i, Y = y_j)}{P(X = x_i)}$$

Property:

- (1) (a) $P(X/Y) \geq 0$ for all real number x
 (b) $P(Y/X) \geq 0$ for all real number y
- (2) (a) $\sum P(X/Y) = 1$
 (b) $\sum P(Y/X) = 1$

Conditional Probability Density Function of X and Y :

Let (X, Y) be a two dimensional random variable of continuous type with joint pdf $f(x, y)$ then the conditional probability density function of X given $Y = y$ is defined as,

$$f(x/y) = \frac{f(x, y)}{f(y)}$$

Here,

$$f(y) = \text{Marginal pdf of } Y = \int_{-\infty}^{\infty} f(x, y) dx$$

Similarly,

the conditional probability density function of Y given $X = x$ is defined as,

$$f(y/x) = \frac{f(x,y)}{f(x)}$$

Here,

$$f(x) = \text{Marginal pdf of } x = \int_{-\infty}^{\infty} f(x,y) dy$$

Q Joint probability density of x and y is

$y \backslash x$	0	1	2
0	0.1	0.4	0.1
1	0.2	0.2	0

Find:

- (i) Probability distribution $P(X=x)$ of x and, probability distribution $P(Y=y)$ of y .
- (ii) Are x and y independent?
- (iii) The conditional probability density of x given $y=1$.
- (iv) Find $F(2,1)$, $F(1,1)$

Solⁿ:

Joint probability density table:

$y \backslash x$	0	1	2	Marginal pdf of $y = P(Y=y)$ to find marginal
0	0.1	0.4	0.1	$P(Y=0) = 0.6$
1	0.2	0.2	0	$P(Y=1) = 0.4$
Marginal pdf of $x = P(X=x)$	$P(X=0) = 0.3$	$P(X=1) = 0.6$	$P(X=2) = 0.1$	1

(i) Prob. distⁿ $P(X=x)$ of x is,

$$P(X=x) = \sum_{y=0,1} P(X=x, Y=y)$$

$$P(X=x) = P(X=x, Y=0) + P(X=x, Y=1)$$

For, $x=0$,

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1)$$

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$$P(X=0) = 0.1 + 0.2 = 0.3$$

For $x=1$,

$$\begin{aligned} P(X=1) &= P(X=1, Y=0) + P(X=1, Y=1) \\ &= 0.4 + 0.2 \\ &= 0.6 \end{aligned}$$

For $x=2$,

$$\begin{aligned} P(X=2) &= P(X=2, Y=0) + P(X=2, Y=1) \\ &= 0.1 + 0 \\ &= 0.1 \end{aligned}$$

Thus, prob. distⁿ $P(X=x)$ of X is,

$P(X=x)$	0.3	0.6	0.1
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(6) prob. distⁿ $P(Y=y)$ of Y is,

$$P(Y=y) = \sum_{x=0,1,2} P(X=x, Y=y)$$

$$\begin{aligned} P(Y=y) &= P(X=0, Y=y) + P(X=1, Y=y) \\ &\quad + P(X=2, Y=y) \end{aligned}$$

For $y=0$,

$$\begin{aligned} P(Y=0) &= P(X=0, Y=0) + P(X=1, Y=0) + P(X=2, Y=0) \\ &= 0.1 + 0.4 + 0.1 \\ &= 0.6 \end{aligned}$$

For $y=1$,

$$\begin{aligned} P(Y=1) &= P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1) \\ &= 0.2 + 0.2 + 0 \\ &= 0.4 \end{aligned}$$

The prob. distⁿ $P(Y=y)$ of Y is,

$P(Y=y)$	0.6	0.4
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(ii) Are X and Y independent?

$$\text{Let } P(X=0) = 0.3$$

$$P(Y=0) = 0.6$$

Also,

$$P(X=0, Y=0) = 0.1$$

Now,

$$P(X=0) \times P(Y=0) = 0.3 \times 0.6 = 0.18$$

$$P(X=0, Y=0) \neq P(X=0) \cdot P(Y=0)$$

$\Rightarrow X$ and Y are dependent.

(iii) The conditional probability density of X given $Y=1$.

we have,

$$f(x/y) = \frac{f(x, y)}{f(y)}$$

$$\Rightarrow f(x/1) = \frac{f(x, 1)}{f(1)} = \frac{P(X=x, Y=1)}{P(Y=1)}$$

$$= \frac{P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1)}{P(Y=1)}$$

$$P\left(\frac{X=0}{Y=1}\right) = \frac{P(X=0, Y=1)}{P(Y=1)}$$

$$\Rightarrow P\left(\frac{X=0}{Y=1}\right) = \frac{0.2}{0.4} = \frac{1}{2}$$

$$\Rightarrow P\left(\frac{X=1}{Y=1}\right) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{0.2}{0.4} = \frac{1}{2}$$

$$\Rightarrow P\left(\frac{X=2}{Y=1}\right) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0}{0.4} = 0$$

$$F(x, y) = P(X \leq x, Y \leq y)$$

(iv) Find $F(2, 1)$, $F(1, 1)$

Here,

$$\begin{aligned} F(2, 1) &= P(X \leq 2, Y \leq 1) \\ &= P(X=0, Y \leq 1) + P(X=1, Y \leq 1) + P(X=2, Y \leq 1) \\ &= P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) \\ &\quad + P(X=1, Y=1) + P(X=2, Y=0) + P(X=2, Y=1) \\ &= 0.1 + 0.2 + 0.4 + 0.2 + 0.1 + 0 \end{aligned}$$

$$\boxed{F(2, 1) = 1}$$

Also,

$$\begin{aligned} F(1, 1) &= P(X \leq 1, Y \leq 1) \\ &= P(X=0, Y \leq 1) + P(X=1, Y \leq 1) \\ &= P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) \\ &\quad + P(X=1, Y=1) \\ &= 0.1 + 0.2 + 0.4 + 0.2 \end{aligned}$$

$$\boxed{F(1, 1) = 0.9}$$

Prob. Distⁿ Function of X and Y :

$F(x, y) = P(X \leq x, Y \leq y)$, if x and y are discrete

$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx$, if x and y are continuous

Properties of prob. distⁿ function:

① $F(-\infty, \infty) = F(-\infty, y) = F(x, -\infty) = 0$

② $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} F(x, y) = 1$

③ $0 \leq F(x, y) \leq 1$

$$(4) P[a_1 < x \leq a_2, b_1 < y \leq b_2] \geq 0$$

(5) If the joint pdf $f(x, y)$ is continuous at (x, y) , then,

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$$

Example:

Practical Throwing a dice two times simultaneously then find its sample space, and also find (i) $F(1, 2)$ (ii) $F(2, 3)$ (iii) $F(6, 6)$

Solⁿ:

Sample space,

$$S = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6) \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6) \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6) \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6) \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6) \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) \end{array} \right\}$$

(i)

$$\text{Also, } F(1, 2) = P(X \leq 1, Y \leq 2)$$

$$= \sum_{x=1}^1 \sum_{y=1}^2 P(X=x, Y=y)$$

$$= \sum_{x=1}^1 [P(X=x, Y=1) + P(X=x, Y=2)]$$

$$= P(X=1, Y=1) + P(X=1, Y=2)$$

$$= \frac{1}{36} + \frac{1}{36}$$

$$= \frac{1}{18}$$

$$\textcircled{17} F(2,3) = P(X \leq 2, Y \leq 3) = \sum_{x=1}^2 \sum_{y=1}^3 P(X \leq x, Y \leq y)$$

$$= \sum_{x=1}^2 \sum_{y=1}^3 P(X \leq 2, Y \leq 3)$$

$$= \sum_{x=1}^2 \left[P(X \leq 2, Y=1) + P(X \leq 2, Y=2) + P(X \leq 2, Y=3) \right]$$

$$= P(X=1, Y=1) + P(X=2, Y=1) + P(X=1, Y=2)$$

$$+ P(X=2, Y=2) + P(X=1, Y=3)$$

$$+ P(X=2, Y=3)$$

$$= \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6} \neq$$

$$\textcircled{17} F(6,6) = P(X \leq 6, Y \leq 6)$$

$$= \sum_{x=1}^6 \sum_{y=1}^6 P(X \leq x, Y \leq y)$$

$$\boxed{F(6,6) = 1}$$

Example: Let (X, Y) be a two joint random variable with joint pdf

$$f(x, y) = kxy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Find (i) value of k (ii) joint prob. distⁿ

function of (X, Y) (iii) $P(X < \frac{1}{2}, Y < \frac{3}{4})$

Solⁿ:

we have,

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x, y) dy \right] dx = 1$$

$$\text{or, } \int_0^1 \left[\int_0^1 f(x,y) dy \right] dx = 1$$

$$\text{or } \int_0^1 \left[\int_0^1 kxy dy \right] dx = 1$$

$$\text{on } \int_0^1 \left[kx \int_0^1 y dy \right] dx = 1$$

$$\text{or, } \int_0^1 \left[kx \left[\frac{y^2}{2} \right]_0^1 \right] dx = 1$$

$$\text{on } \int_0^1 \frac{1}{2} kx dx = 1$$

$$\text{or, } \frac{1}{2} k \int_0^1 x dx = 1$$

$$\text{on } \frac{1}{2} k \left[\frac{x^2}{2} \right]_0^1 = 1$$

$$\text{on } \frac{1}{2} k \times \frac{1}{2} = 1$$

$$\text{on } \frac{k}{4} = 1$$

$$\Rightarrow k = 4$$

Now, the function becomes,

$$f(x,y) = 4xy$$

(ii) The joint probability distribution function of (X,Y) is given by,

$$F(x,y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x,y) dx dy$$

$$F(x,y) = \int_0^x \int_0^y 4xy dx dy$$

$$= \int_0^y \left[4x \int_0^x y dx \right] dy$$

$$= \int_0^y 4y \frac{x^2}{2} dy$$

$$= \int_0^x 2xy^2 dx$$

$$= \frac{2}{2} y^2 \int_0^x x dx$$

$$= y^2 \cdot \frac{x^2}{2}$$

$$\boxed{F(x, y) = x^2 y^2}$$

$$(iii) P\left(x < \frac{1}{2}, y < \frac{3}{4}\right) = \int_0^{1/2} \int_0^{3/4} f(x, y) dx dy$$

$$= \int_0^{1/2} \int_0^{3/4} 4xy dx dy$$

$$= \int_0^{1/2} \left[4x \int_0^{3/4} y dy \right] dx$$

$$= \int_0^{1/2} 4x \cdot \left[\frac{y^2}{2} \right]_0^{3/4} dx$$

$$= \int_0^{1/2} 4x \cdot \frac{1}{2} \times \frac{9}{16} dx$$

$$= \int_0^{1/2} \frac{9}{8} x dx$$

$$= \frac{9}{8} \int_0^{1/2} x dx$$

$$= \frac{9}{8} \left[\frac{x^2}{2} \right]_0^{1/2}$$

$$= \frac{9}{16} \times \frac{1}{4}$$

$$\boxed{P(x < 1/2, y < 3/4) = \frac{9}{64}}$$

Marginal pmf of x and y :

(a) Marginal pmf of x , $P(x=x) = \sum_y P(x=x, y=y)$

(b) Marginal pmf of y , $P(y=y) = \sum_x P(x=x, y=y)$

Example: Let X and Y be a discrete random variable and the joint pmf is,

$$P(X=x, Y=y) = K(2x+y) \quad ; \quad x=0,1,2 \\ y=0,1,2,3$$

Find (i) value of K

(ii) the marginal pmf of X and Y

(iii) $P(X \geq 1, Y \leq 2)$

(iv) $P(X+Y \leq 3)$

Solⁿ:

$$K = \frac{1}{42}$$

(i) we have,

$$\sum_{x=0}^2 \sum_{y=0}^3 P(X=x, Y=y) = 1$$

$$\Rightarrow \sum_{x=0}^2 [P(X=x, Y=0) + P(X=x, Y=1) + P(X=x, Y=2) + P(X=x, Y=3)] = 1$$

$$\Rightarrow P(X=0, Y=0) + P(X=1, Y=0) + P(X=2, Y=0) + P(X=0, Y=1)$$

$$+ P(X=1, Y=1) + P(X=2, Y=1) + P(X=0, Y=2) + P(X=1, Y=2)$$

$$+ P(X=2, Y=2) + P(X=0, Y=3) + P(X=1, Y=3) + P(X=2, Y=3) = 1$$

$$\Rightarrow 0 + 2K + 4K + K + 3K + 5K + 2K + 4K + 6K + 3K + 5K + 7K = 1$$

$$\Rightarrow 42K = 1$$

$$\Rightarrow K = \frac{1}{42}$$

(ii) Marginal pmf of X .

$$P(X=x) = \sum_{y=0}^3 P(X=x, Y=y)$$

$$P(X=x) = P(X=x, Y=0) + P(X=x, Y=1) + P(X=x, Y=2) + P(X=x, Y=3)$$

For $x=0$,

$$P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=0, Y=2) + P(X=0, Y=3)$$

Table

$y \backslash x$	0	1	2	marginal prob of y
0	0	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{1}{7}$
1	$\frac{1}{42}$	$\frac{1}{14}$	$\frac{5}{42}$	$\frac{3}{14}$
2	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{1}{7}$	$\frac{2}{7}$
3	$\frac{1}{14}$	$\frac{5}{42}$	$\frac{1}{6}$	$\frac{5}{14}$
marginal prob of x	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{2}{21}$	1

$$= 0 + \frac{1}{42} + \frac{1}{21} + \frac{1}{14}$$

$$= \frac{1}{14}$$

For $x=1$,

$$P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) + P(X=1, Y=2) + P(X=1, Y=3)$$

$$= \frac{1}{21} + \frac{1}{14} + \frac{2}{21} + \frac{5}{42}$$

$$= \frac{1}{3}$$

For $x=2$,

$$P(X=2) = P(X=2, Y=0) + P(X=2, Y=1) + P(X=2, Y=2) + P(X=2, Y=3)$$

$$= \frac{2}{21} + \frac{5}{42} + \frac{1}{7} + \frac{1}{6}$$

$$= \frac{2}{21}$$

The marginal prob of x is,

$P(X=x)$	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{2}{21}$
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[Ans]

The marginal prob of x is given by

$$P(X=x) = \sum_y P(X=x, Y=y) = \sum_{y=0}^3 P(X=x, Y=y)$$

$$P(X=x) = \frac{1}{42} \sum_{y=0}^3 (2x+y) = \frac{1}{42} (0+y + 2+y + 4+y + 6+y)$$

$$= \frac{1}{42} (3y+12) = \frac{1}{42} 3(y+4) = \frac{1}{14} (y+4)$$

$$1 \cdot x + 3$$

$$x = 0, 1, 2$$

② marginal prob of x ,

$$P(Y=y) = \sum_{x=0}^2 P(X=x, Y=y)$$

$$P(Y=y) = P(X=0, Y=y) + P(X=1, Y=y) + P(X=2, Y=y)$$

For $y=0$,

$$\begin{aligned} P(Y=0) &= P(X=0, Y=0) + P(X=1, Y=0) + P(X=2, Y=0) \\ &= 0 + \frac{1}{14} + \frac{2}{14} \\ &= \frac{1}{7} \end{aligned}$$

For $y=1$,

$$\begin{aligned} P(Y=1) &= P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1) \\ &= \frac{1}{14} + \frac{1}{14} + \frac{5}{14} \\ &= \frac{3}{7} \end{aligned}$$

For $y=2$,

$$\begin{aligned} P(Y=2) &= P(X=0, Y=2) + P(X=1, Y=2) + P(X=2, Y=2) \\ &= \frac{1}{14} + \frac{2}{14} + \frac{1}{7} \\ &= \frac{2}{7} \end{aligned}$$

For $y=3$,

$$\begin{aligned} P(Y=3) &= P(X=0, Y=3) + P(X=1, Y=3) + P(X=2, Y=3) \\ &= \frac{1}{14} + \frac{5}{14} + \frac{1}{6} \\ &= \frac{5}{14} \end{aligned}$$

The marginal prob of y is,

$P(Y=y)$	$\frac{1}{7}$	$\frac{3}{7}$	$\frac{2}{7}$	$\frac{5}{14}$
----------	---------------	---------------	---------------	----------------

The marginal prob of x is given by,

$$\begin{aligned} P(X=x) &= \sum_{y=0}^3 P(X=x, Y=y) = \frac{1}{14} \sum_{y=0}^3 (2x+y) \\ &= \frac{y+2}{14} \quad | \quad y=0,1,2,3 \end{aligned}$$

$$(iii) P(X \geq 1, Y \leq 2)$$

$$= P(X \geq 1, Y=0) + P(X \geq 1, Y=1) + P(X \geq 1, Y=2)$$

$$= P(X=1, Y=0) + P(X=2, Y=0) + P(X=1, Y=1) + P(X=2, Y=1) \\ + P(X=1, Y=2) + P(X=2, Y=2)$$

$$= \frac{1}{12} + \frac{2}{24} + \frac{1}{14} + \frac{5}{42} + \frac{2}{24} + \frac{1}{7}$$

$$= \frac{4}{7}$$

$$(iv) P(X+Y < 3) = \sum_{x \leq 2} \sum_{y=0}^{2-x} P(X=x, Y=y)$$

$$= P(X \leq 2, Y=2-x)$$

$$= P(X=0, Y=2) + P(X=1, Y=1) + P(X=2, Y=0)$$

$$= \frac{1}{24} + \frac{1}{14} + \frac{2}{24}$$

$$= \frac{3}{14}$$

$$(v) P(X \geq 1, Y \leq 2) = \sum_{x \geq 1} \sum_{y \leq 2} P(X=x, Y=y)$$

$$= \frac{1}{42} \sum_{x \geq 1} \sum_{y \leq 2} (2x+y)$$

$$= \frac{1}{42} \left[\sum_{x \geq 1} 2x \left(\sum_{y \leq 2} 1 \right) + \sum_{y \leq 2} y \left(\sum_{x \geq 1} 1 \right) \right]$$

$$= \frac{1}{42} \left[(2+4)(3) + (0+1+2)(2) \right]$$

$$= \frac{1}{42} (5+6) = \frac{11}{42}$$

$$(vi) P(X+Y < 3) = \sum_{x \leq 2} \sum_{y=0}^{2-x} P(X=x, Y=y)$$

$$= \frac{1}{42} \sum_{x=0,1,2} \sum_{y=0,1,2-x} (2x+y)$$

$$= \frac{1}{42} \left[(0+0)(0+1+2) + (0+1)(0+1) + (2+0) + (2+1) + (4+0) \right]$$

$$= \frac{12}{42}$$

Marginal pdf of X and Y:

Let (X, Y) be a two dimensional random variable of continuous type with joint pdf $f(x, y)$ then the marginal pdf of X denoted by $f(x)$, and that of Y denoted by $f(y)$ respectively are defined by,

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Example: let (X, Y) be jointly distributed random variable with pdf given by,

$$f(x, y) = 24xy \quad ; \quad \begin{array}{l} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 \\ x+y \leq 1 \end{array}$$

= 0, otherwise

solⁿ: Find (i) $f(x)$, (ii) $f(y)$, (iii) $P(X+Y < 0.5)$

Here,

$$f(x, y) = 24xy$$

$$\begin{aligned} \textcircled{i} f(x) &= \int_0^{1-x} 24xy dy = 24x \int_0^{1-x} y dy \\ &= 24x \left[\frac{y^2}{2} \right]_0^{1-x} \\ &= \frac{24x}{2} (1-x)^2 \\ &= 12x(1-x)^2 \end{aligned}$$

$$\begin{aligned} \textcircled{ii} f(y) &= \int_0^{1-y} 24xy dx = 24y \int_0^{1-y} x dx \\ &= 24y \left[\frac{x^2}{2} \right]_0^{1-y} \\ &= \frac{24y}{2} [(1-y)^2 - 0] \\ &= 12y(1-y)^2 \end{aligned}$$

$$(10) P(x+y < 0.5) = \int_0^{0.5} \int_0^{0.5-x} f(x,y) dx dy$$

$$= \int_0^{0.5} \int_0^{0.5-x} 24xy dx dy$$

$$= 24 \int_0^{0.5} x \left[\int_0^{0.5-x} y dy \right] dx$$

$$= 24 \int_0^{0.5} x \left[\left[\frac{y^2}{2} \right]_0^{0.5-x} \right] dx$$

$$= \frac{24}{2} \int_0^{0.5} x (0.5-x)^2 dx$$

$$= 12 \int_0^{0.5} x \left(\frac{1}{4} - x + x^2 \right) dx$$

$$= 12 \int_0^{0.5} \left(\frac{1}{4}x - x^2 + x^3 \right) dx$$

$$= 12 \left[\frac{x^2}{8} - \frac{x^3}{3} + \frac{x^4}{4} \right]$$

$$= 12 \left[\frac{1}{4} \int_0^{0.5} x dx + \int_0^{0.5} x^2 dx + \int_0^{0.5} x^3 dx \right]$$

$$= 12 \left[\frac{1}{4} \times 0.5 - \frac{1}{2} \times \frac{1}{4} + \frac{1}{4} \right]$$

$$= 12 \left[\frac{1}{4} \left[\frac{x^2}{2} \right]_0^{0.5} - \left[\frac{x^3}{3} \right]_0^{0.5} + \left[\frac{x^4}{4} \right]_0^{0.5} \right]$$

$$= 12 \left[\frac{1}{8} \times 0.5^2 - \frac{1}{3} \times 0.5^3 + \frac{1}{4} \times 0.5^4 \right]$$

$$= 12 \left[\frac{1}{32} - \frac{1}{24} + \frac{1}{64} \right]$$

$$= \frac{1}{16}$$

$$= 0.0625$$

Independent of Random Variable:

Theorem: Let (x, y) be a two dimensional continuous random variable with joint pdf $f(x, y)$ and $f(x)$ and $f(y)$ be the marginal pdf of x and y respectively. Then x and y are said to be independent if and only if $f(x, y) = f(x) \times f(y)$

Proof:

Let (x, y) be a continuous random variable.

Necessary condition: If x and y are independent random variables then $f(x, y) = f(x) \cdot f(y)$.

where, $f(x, y)$ is joint pdf of x & y

$f(x)$ is marginal pdf of x

$f(y)$ is marginal pdf of y

Thus the necessary condition is satisfied.

Sufficient Condition: If $f(x, y) = g(x) \cdot g(y)$ then we have to prove that x and y are independent. The marginal pdf of x and y respectively are given by,

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_{-\infty}^{\infty} g(x) \cdot h(y) dy$$

$$f(x) = g(x) \int_{-\infty}^{\infty} h(y) dy$$

$$\boxed{f(x) = g(x) \times k_1}$$

$$\text{where } k_1 = \int_{-\infty}^{\infty} h(y) dy$$

and,

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{-\infty}^{\infty} g(x) \cdot h(y) dx$$

$$f(y) = h(y) \int_{-\infty}^{\infty} g(x) dx = h(y) \times k_2$$

$$\boxed{f(y) = k_2 h(y)}$$

we know,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x) \cdot h(y) dx dy = 1$$

$$\Rightarrow \left[\int_{-\infty}^{\infty} g(x) dx \right] \left[\int_{-\infty}^{\infty} h(y) dy \right] = 1$$

$$K_1 \cdot K_2 = 1$$

$$\therefore f(x, y) = g(x) \cdot h(y)$$

$$= K_1 g(x) \cdot K_2 h(y)$$

$$\boxed{\therefore f(x, y) = f(x) \cdot f(y)}$$

This implies that ~~is~~ X and Y are independent.

Ex

Example: The joint prob. distⁿ of X and Y is given by,

$$f(x, y) = 4xy e^{-(x^2+y^2)} \quad ; \quad x \geq 0 ; y \geq 0$$

- (a) Show that X and Y are independent
(b) Find the conditional distribution of X for given $Y=y$.

Solution:

(a) we know,

The marginal pdf of X is given by,

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} 4xy e^{-x^2} \cdot e^{-y^2} dy$$

$$f(x) = 4x e^{-x^2} \int_0^{\infty} y e^{-y^2} dy$$

$$\text{Put } y^2 = t \Rightarrow y = t^{1/2}$$

$$2y dy = dt$$

$$dy = \frac{dt}{2\sqrt{t}} = \frac{dt}{2t^{1/2}}$$

$$f(x) = \frac{2}{4\pi} e^{-x^2} \int_0^\infty t^{1/2} e^{-t} \frac{dt}{2t^{1/2}}$$

$$= 2\pi e^{-x^2} \int_0^\infty e^{-t} dt$$

$$= 2\pi e^{-x^2} \left[\frac{e^{-t}}{-1} \right]_0^\infty$$

$$= 2\pi e^{-x^2} \left[-e^{-\infty} + e^{-0} \right] \Rightarrow 1$$

$$f(x) = 2\pi e^{-x^2}$$

Similarly, the marginal pdf of y is,

$$f(y) = 2\pi e^{-y^2}$$

Now,

$$f(x) \cdot f(y) = 2\pi e^{-x^2} \cdot 2\pi e^{-y^2}$$

$$= 4\pi^2 e^{-(x^2+y^2)}$$

$$= f(x, y)$$

$\therefore X$ and Y are independent.

(b) The conditional distribution of x given $y=y$ is,

$$f(x|y) = \frac{f(x, y)}{f(y)} = \frac{4\pi^2 e^{-x^2} \cdot e^{-y^2}}{2\pi e^{-y^2}}$$

$$= 2\pi e^{-x^2}$$

$$\therefore f(x|y) = 2\pi e^{-x^2}, x \geq 0$$

Example: The joint pdf of two continuous random variable x and y is $f(x, y) = kxy$, $0 < x < 5$

Find (i) $k = ?$

(ii) $f(x) = ?$, $f(y) = ?$

(iii) $f(y/x) = ?$, $f(x/y) = ?$

(iv) Check whether two variable x and y are independent or not?

(v) $P(0 < x < 2, 1 < y < 3)$

(vi) $P(x < 3 / y > 2)$

Solⁿ:

(i)

We have,

$k = \frac{8}{625}$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\Rightarrow \int_0^5 \left[\int_0^5 kxy dy \right] dx = 1$$

$$\Rightarrow \int_0^5 \frac{kx}{2} (5^2 - x^2) dx = 1$$

$$\Rightarrow \int_0^5 \left(\frac{25kx}{2} - \frac{kx^3}{2} \right) dx = 1$$

$$\Rightarrow \frac{25k}{2} \int_0^5 x dx - \frac{k}{2} \int_0^5 x^3 dx = 1$$

$$\Rightarrow \frac{25k}{2} \times \frac{1}{2} \cdot 5^2 - \frac{k}{2} \cdot \frac{1}{4} \cdot 5^4 = 1$$

$$\Rightarrow \frac{625k}{4} - \frac{625k}{8} = 1$$

$$\Rightarrow \frac{625}{8} k = 1$$

$$\Rightarrow k = \frac{8}{625}$$

(1) $f(x) = ?$

we have,

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^5 \frac{8}{625} xy dy$$

$$= \frac{8}{625} x \cdot \int_x^5 y dy$$

$$= \frac{8}{625} x \cdot \frac{1}{2} (5^2 - x^2)$$

$$= \frac{4}{625} x \cdot 25 - \frac{4}{625} x \cdot x^2$$

$$= \frac{4}{25} x - \frac{4}{625} x^3$$

$$= \frac{4}{625} x (25 - x^2)$$

Similarly,

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y \frac{8}{625} xy dx$$

$$= \int_0^y \frac{8}{625} y \int_0^y x dx$$

$$= \frac{8}{625} y \cdot \frac{1}{2} (y^2 - 0)$$

$$= \frac{4}{625} y^3$$

$$(ii) f(y/x) = \frac{f(x, y)}{f(x)} = \frac{\frac{8}{625} xy}{\frac{4}{625} x (25 - x^2)} = \frac{2y}{25 - x^2}$$

$$f(x/y) = \frac{f(x, y)}{f(y)} = \frac{\frac{8}{625} xy}{\frac{4}{625} y^3} = \frac{2x}{y^2}$$

(40) $\Rightarrow$

$$f(x) \times f(y) = \frac{64}{625} x(25-x^2) \times \frac{4}{625} y^3$$

$$= \frac{16}{390625} xy^3(25-x^2) \quad \text{--- (1)}$$

$$p(x, y) = \frac{8}{625} xy \quad \text{--- (11)}$$

From (1) and (11),

$$f(x, y) \neq f(x) \cdot f(y)$$

 $\therefore x$ and y are dependent.

$$(2) P(0 < x < 2, 1 < y < 3) = \int_0^2 \int_1^3 \frac{8xy}{625} dx dy$$

$$= \int_0^2 \left[\frac{8}{625} x \int_1^3 y dy \right] dx$$

$$= \int_0^2 \left[\frac{8}{625} x \left(\frac{y^2}{2} \right) \right]_1^3 dx$$

$$= \int_0^2 \frac{8}{625} x \cdot \frac{1}{2} (3^2 - 1) dx$$

$$= \int_0^2 \frac{32}{625} x dx$$

$$= \frac{32}{625} \int_0^2 x dx = \frac{32}{625} \left(\frac{x^2}{2} \right) \Big|_0^2$$

$$= \frac{16}{625} \cdot \frac{32}{2} \cdot 1 \cdot 2^2$$

$$= \frac{64}{625}$$

$$(vi) P(X < 3 / Y > 2) = \frac{P(X < 3, Y > 2)}{P(Y > 2)}$$

where,

$$P(X < 3, Y > 2) = \int_0^3 \int_2^5 \frac{8}{625} xy \, dx \, dy$$

$$= \int_0^3 \left[\frac{8}{625} x \int_2^5 y \, dy \right] dx$$

$$= \int_0^3 \frac{8}{625} x \cdot \frac{1}{2} (5^2 - 2^2) \, dx$$

$$= \int_0^3 \frac{8}{625} \cdot \frac{1}{2} \cdot 21 \, dx$$

$$= \frac{84}{625} \int_0^3 x \, dx$$

$$= \frac{84}{625} \cdot \frac{1}{2} \cdot 3^2$$

$$= \frac{378}{625}$$

also, $P(Y > 2) = \int_2^5 f(y) \, dy$

$$= \int_2^5 \frac{4}{625} y^3 \, dy$$

$$= \frac{4}{625} \int_2^5 y^3 \, dy$$

$$= \frac{4}{625} \cdot \frac{1}{4} (5^4 - 2^4)$$

$$= \frac{609}{625}$$

now,

$$P(X < 3 / Y > 2) = \frac{P(X < 3, Y > 2)}{P(Y > 2)} = \frac{\frac{378}{625}}{\frac{609}{625}} = \frac{18}{29}$$

Transformation of two dimensional Random Variable:

The Jacobian of the transformation of (X, Y) into (U, V) is defined as a determinant of the partial derivatives of x and y with respect to U and V and it is denoted by J .

Mathematically,

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}$$

The Jacobian of transformation J plays an important role in the determination of joint pdf of the new random variable (U, V) from the joint pdf of the old random variable (X, Y) .

Distribution of Sum of two Random Variables:

If X and Y are two continuous random variables with joint pdf $f(x, y)$ then the distribution of $U = X + Y$ is given by,

$$g(u) = \int_{-\infty}^{\infty} f(v, u-v) dv \quad ; -\infty < u < \infty$$

$$g(u) = \int_{-\infty}^{\infty} f(v) f(u-v) dv \quad ; \text{if } X \text{ and } Y \text{ are independent}$$

Proof:

To find the distribution of sum $X + Y$, we make a transformation as

$$\begin{aligned} u &= x + y & \Rightarrow y &= u - x = u - v \\ v &= x & \Rightarrow x &= v \end{aligned}$$

The Jacobian of transformation is given by,

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

$$|J| = 1$$

Then, the joint pdf of the two new random variables u and v is given by,

$$g(u, v) = f(x, y) |J|$$

$$g(u, v) = f(v, u-v) \quad ; -\infty < u < \infty$$

$$-\infty < v < \infty$$

The marginal pdf of u is given as,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv = \int_{-\infty}^{\infty} f(v, u-v) dv$$

$$= \int_{-\infty}^{\infty} f(v) \cdot f(u-v) dv$$

if x and y are independent.

If x and y are two non-negative independent random variables with the respective pdf $f(x)$ and $f(y)$ then

$$g(u) = \int_0^u f(v) \cdot f(u-v) dv$$

Example: Let x and y be the lengths of two electronic devices and have joint pdf given by,

$$f(x, y) = e^{-(x+y)} \quad ; 0 \leq x, 0 \leq y$$

$$= 0 \quad ; \text{otherwise}$$

Find the marginal pdf for $U = X + Y$.
Also, find the mean and variance.

Solution:

Let us make transformation,

$$U = X + Y \quad \text{and} \quad V = X$$

$$\Rightarrow X = V \quad \text{and} \quad Y = U - V$$

Then the Jacobian of transformation is given by,

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix}$$

$$J = -1$$

$$|J| = 1$$

Thus, the joint pdf of U and V is given by,

$$g(u, v) = f(x, y) |J|$$

$$= e^{-(x+y)} = 1$$

$$g(u, v) = e^{-u} \quad ; \quad 0 \leq u, \quad 0 \leq v \leq u$$

$$= 0 \quad , \quad \text{otherwise}$$

The marginal pdf of U is given by,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv$$

$$= \int_0^u e^{-u} dv$$

$$= \int_0^u e^{-u} [v]_0^u$$

$$= u e^{-u} \quad ; \quad 0 \leq u$$

$$0 \quad ; \quad \text{otherwise}$$

The mean of U is given by,

$$E(U) = \int_{-\infty}^{\infty} u g(u) du$$

$$= \int_0^{\infty} u \cdot u e^{-u} du$$

$$= \int_0^{\infty} u^2 e^{-u} du$$

$$= \int_0^{\infty} e^{-u} u^{3-2} du$$

$$= \sqrt{3}$$

$$= 2!$$

$$= 2$$

$$\boxed{E(U) = 2}$$

Again,

$$E(U^2) = \int_{-\infty}^{\infty} u^2 g(u) du$$

$$= \int_0^{\infty} u^2 u e^{-u} du$$

$$= \int_0^{\infty} e^{-u} u^3 du$$

$$= \int_0^{\infty} e^{-u} u^{4-1} du$$

$$= \sqrt{4}$$

$$= 3!$$

$$E(U^2) = 6$$

$$\therefore V(U) = E(U^2) - [E(U)]^2$$

$$= 6 - 2^2$$

$$= 2$$

$$\boxed{\therefore V(U) = 2}$$

Distribution of product of two random variable:

Let X and Y be two continuous random variable with pdf $f(x)$ and $f(y)$ respectively then pdf of $U = XY$ is given by,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv$$

$$= \int_{-\infty}^{\infty} f(v, y/v) \frac{1}{|v|} dv$$

If X and Y are independent,

$$g(u) = \int_{-\infty}^{\infty} f(v) \cdot f(y/v) \frac{1}{|v|} dv$$

Proof:

Let $f(x, y)$ be a joint pdf of random variables X and Y . Let us make a transformation $u = xy$ and $v = x$.

$$\Rightarrow y = u/v \quad x = v$$

Then, the Jacobian of transformation is given by,

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & \frac{1}{v} \\ 1 & -\frac{1}{v^2} \end{vmatrix} = \frac{1}{v}$$

The joint pdf of random variables u and v is given by,

$$g(u, v) = f(x, y) |J| = f(v, y/v) \frac{1}{|v|}$$

Hence, the marginal pdf of U is given by,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv = \int_{-\infty}^{\infty} f(v, y/v) \frac{1}{|v|} dv$$

For independent of random variables $X \in Y$,

Prob. density function of U is,

$$g(u) = \int_{-\infty}^{\infty} f(v) \cdot f(y/v) \frac{1}{|v|} dv$$

Example: The joint pdf of two random variables X and Y is given by $f(x, y) = 2(x+y)$, $0 < x < 1$, $0 < y < 1$. Find the pdf of XY . Also, find the mean and variance of the transformed distribution of XY .

Soln:

Let us make transformation,

$$u = xy \quad \text{and} \quad v = x$$

$$\Rightarrow x = v \quad , \quad y = u/v$$

The Jacobian of transformation is,

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1/v \\ 1 & -u/v^2 \end{vmatrix} = -\frac{1}{v}$$

$$|J| = \frac{1}{v}$$

Then, the joint pdf of u and v is given by,

$$\begin{aligned} g(u, v) &= f(x, y) |J| \\ &= (x+y) |J| \\ &= \left(v + u/v\right) \frac{1}{v} \end{aligned}$$

$$\boxed{g(u, v) = \left(1 + \frac{u}{v^2}\right)}$$

The marginal pdf of U is,

$$\begin{aligned} g(u) &= \int_{-\infty}^{\infty} g(u, v) dv \\ &= \int_0^1 \left(1 + \frac{u}{v^2}\right) dv \end{aligned}$$

$$\boxed{g(u) = 2(1-u)} \quad , \quad 0 < u < 1$$

The mean of the transformed distribution of $U = XY$ is given by,

$$E(U) = \int_0^1 u g(u) du$$

$$= \int_0^1 u \cdot 2(1-u) du$$

$$= 2 \int_0^1 (u - u^2) du = \int_0^1 4u du - \int_0^1 4u^2 du$$

$$= \frac{4}{2} - \frac{4}{3} \cdot 1$$

$$= \frac{4}{3}$$

$$= \frac{3-2}{6} = \frac{1}{6}$$

and, $E(U^2) = \int_0^1 u^2 g(u) du$

$$= \int_0^1 u^2 \cdot 2(1-u) du$$

$$= \frac{2}{6} \int_0^1 u^2 du - \int_0^1 4u^3 du = \frac{1}{12}$$

$$\therefore V(U) = E(U^2) - [E(U)]^2$$

$$= \frac{1}{12} - \left(\frac{1}{6}\right)^2 = \frac{1}{12} - \left(\frac{1}{6}\right)^2$$

$$\boxed{V(U) = 0.0555} = \frac{1}{18}$$

$$\therefore V(U) = 0.0555 \neq$$

Let (X, Y) be a random variable of continuous type with pdf $f(x, y)$. Then the pdf for $U = \frac{X}{Y}$ is given by

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv = \int_{-\infty}^{\infty} f(uv, v) |v| dv$$

If x and y are independent random variables with pdf $f(x)$ and $f(y)$ then

$$g(u) = \int_{-\infty}^{\infty} f(uv) \cdot f(v) |v| dv$$

Proof:

Let $f(x, y)$ be joint pdf of x and y then let us make the transformation

$$U = \frac{x}{y} \quad \text{and} \quad v = y$$

$$\Rightarrow x = uv \quad \text{and} \quad y = v$$

Then the Jacobian of transformation is given by,

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v$$

Thus the joint pdf of U and V is,

$$\begin{aligned} g(u, v) &= f(x, y) |J| \\ &= f(uv, v) |v| \end{aligned}$$

Hence the marginal pdf of U is given by,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv = \int_{-\infty}^{\infty} f(uv, v) |v| dv$$

If x and y are independent variables with pdf $f(x)$ and $f(y)$,

$$g(u) = \int_{-\infty}^{\infty} f(uv) f(v) |v| dv$$

Example: Let x_1 and x_2 be two independent random variables with pdf,

$$f(x) = e^{-x} \quad , x > 0$$

$$= 0 \quad , \text{otherwise}$$

Find the pdf of $U = x_1 + x_2$ and

$$V = \frac{x_1}{x_1 + x_2} \quad \text{and show that } U \text{ and } V$$

are independent

Solⁿ:

Since x_1 and x_2 are two independent random variables with common pdf $f(x) = e^{-x}$, $x > 0$ then the joint pdf of x_1 and x_2 is given by,

$$f(x_1, x_2) = f(x_1) \cdot f(x_2) = e^{-(x_1 + x_2)}, \quad x_1 > 0, \quad x_2 > 0$$

To find the pdf of $U = x_1 + x_2$ and $V = \frac{x_1}{x_1 + x_2}$.

Let us make a transformation,

$$x_1 = uv \quad \text{and} \quad x_2 = u(1-v)$$

Then the Jacobian of transformation is given by,

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} = \begin{vmatrix} v & 1-v \\ u & -u \end{vmatrix}$$

$$= -uv - u + uv$$

$$|J| = |-u| = u$$

Thus the joint pdf of U and V is,

$$g(u, v) = f(x_1, x_2) |J|$$

$$= f(uv, u(1-v)) \cdot u$$

$$g(u, v) = e^{-(u+v)} u \quad , u > 0 \text{ and } 0 < v < 1$$

$$= \begin{cases} ue^{-u} & , u > 0 \text{ and } 0 < v < 1 \\ 0 & \text{otherwise} \end{cases}$$

Now,

the marginal pdf of U is given by,

$$g(u) = \int_{-\infty}^{\infty} g(u, v) dv$$

$$= \int_0^1 ue^{-u} dv \quad , 0 < v < 1, u > 0$$

$$= \int_0^1 ue^{-u} dv$$

$$= ue^{-u} [v]_0^1$$

$$= \begin{cases} ue^{-u} & ; u > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

Similarly,

the marginal pdf of V is given by,

$$g(v) = \int_{-\infty}^{\infty} g(u, v) du$$

$$= \int_0^{\infty} ue^{-u} du \quad ; u > 0, 0 < v < 1$$

$$= \int_0^{\infty} ue^{-u} du$$

$$= \int_0^{\infty} e^{-u} u^2 du$$

$$= \sqrt{2}$$

$$= 1$$

$$\text{Since, } g(u, v) = ue^{-u}$$

$$\therefore g(u) = ue^{-u} \text{ and } g(v) = 1$$

$$\therefore g(u) \cdot g(v) = ue^{-u} \cdot 1 = ue^{-u}$$

$$\Rightarrow g(u, v) = g(u) \cdot g(v) = ue^{-u}$$

Then, these two new random variables

$U = X_1 + X_2$ and $V = \frac{X_1}{X_1 + X_2}$ are independent.

Example: Let the pdf of the random variable (X, Y) is,

$$f(x, y) = \frac{1}{\alpha^2} e^{-(x+y)/\alpha}, \quad x > 0, y > 0, \alpha > 0$$

Find the distribution of $\frac{1}{2}(X-Y)$.

Soln:

$$\text{Let } u = \frac{1}{2}(x-y) \text{ and } v = y$$

$$\Rightarrow y = v \quad \text{and } x = 2u + v$$

The Jacobian of transformation is given by,

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2$$

Thus, the joint pdf of U and V is,

$$g(u, v) = f(x, y) |J|$$

$$g(u, v) = \frac{2}{\alpha^2} e^{-(2u+2v)/\alpha}$$

Domain of u, v

$$\text{when } x > 0 \Rightarrow 2u + v \quad \text{i.e., } v > -2u$$

$$\text{when } y > 0 \Rightarrow v > 0$$

Thus, the required pdf of U and V is,

$$g(u, v) = \frac{2}{\alpha^2} e^{-(2u+2v)/\alpha} \quad ; -\infty < u < \infty$$

$$; v > 0, v > -2u$$

To find the distribution of U :

The marginal pdf of U is,

$$g(u) = \int \frac{2}{\alpha^2} e^{-(2u+2v)/\alpha} dv$$

$$g(u) = \frac{2}{\alpha^2} e^{-2u/\alpha} \int e^{-2v/\alpha} dv$$

when $-\infty < u < 0$,

$$\begin{aligned} g(u) &= \frac{2}{\alpha^2} e^{-2u/\alpha} \int_{-\infty}^{\infty} e^{-2v/\alpha} dv \\ &= \frac{2}{\alpha^2} e^{-2u/\alpha} \left[\frac{e^{-2v/\alpha}}{-2/\alpha} \right]_{-\infty}^{\infty} \\ &= \frac{1}{\alpha} e^{2u/\alpha} \end{aligned}$$

when $0 < u < \infty$

$$\begin{aligned} g(u) &= \frac{2}{\alpha^2} e^{-2u/\alpha} \int_0^{\infty} e^{-2v/\alpha} dv \\ &= \frac{2}{\alpha^2} e^{-2u/\alpha} \left[\frac{e^{-2v/\alpha}}{-2/\alpha} \right]_0^{\infty} \\ &= \frac{1}{\alpha} e^{-2u/\alpha}, \quad 0 < u < \infty \end{aligned}$$

$$\therefore g(u) = \frac{1}{\alpha} e^{\frac{2}{\alpha} u}, \quad -\infty < u < \infty$$

Example: The joint density function of two random variables x_1 and x_2 is $f(x_1, x_2) = 2e^{-x_1-x_2}$; $0 < x_1 < x_2 < \infty$.

Consider the transformation $U = 2x_1$; $V = x_2 - x_1$. Find the joint density function of U and V . Check whether U and V are independent?

Soln:

Given,

$$U = 2x_1$$

$$V = x_2 - x_1$$

$$\Rightarrow x_1 = U/2$$

$$\Rightarrow x_2 = V + U/2$$

The Jacobian of transformation is,

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{vmatrix} = \frac{1}{2}$$

Thus, the pdf of U and V is given by,

$$g(u, v) = f(x, y) |J|$$

$$= 2e^{-(u+v)} \times \frac{1}{2}$$

$$g(u, v) = e^{-(u+v)}$$

For limits,

$$\text{when } 0 < x_1 < x_2 < \infty$$

$$0 < u/2 < v + u/2 < \infty$$

$$\Rightarrow u > 0, \quad 0 < v < \infty$$

The marginal pdf of U is,

$$g(u) = \int_0^{\infty} e^{-u} e^{-v} dv$$

$$= e^{-u} \int_0^{\infty} e^{-v} dv$$

$$= e^{-u}$$

The marginal pdf of V is,

$$g(v) = \int_0^{\infty} e^{-u} e^{-v} du$$

$$= e^{-v} \int_0^{\infty} e^{-u} du$$

$$= e^{-v}$$

$$g(u, v) = g(u) \cdot g(v) = e^{-u} \cdot e^{-v} = e^{-(u+v)}$$

Thus, they are independent.

Example: The joint pdf of two random variables x_1 and x_2 is $f(x_1, x_2) = 8x_1x_2$, $0 < x_1 < x_2 < 1$. Consider the transformation $U = x_1/x_2$; $V = x_2$. Find the joint pdf of U and V . Check whether U and V are independent or not?

Solⁿ:

Given that,

$$U = x_1/x_2$$

$$V = x_2$$

$$\Rightarrow x_1 = UV$$

$$\Rightarrow x_2 = V$$

The Jacobian of transformation is,

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} = \begin{vmatrix} V & U \\ 0 & 1 \end{vmatrix} = V$$

Thus, the pdf of U and V is,

$$\begin{aligned} g(u, v) &= f(x_1, x_2) |J| \\ &= 8(uv)(v) \cdot v \\ &= 8uv^3 \end{aligned}$$

For limits:

$$0 < x_1 < x_2 < 1$$

$$0 < UV < V < 1$$

$$UV > 0 \quad V < 1$$

$$U(1-V) < 0$$

Thus, domain of U, V are

$$0 < U < 1, \quad 0 < V < 1$$

$$\Rightarrow g(u, v) = 8uv^3 \quad ; \quad 0 < U < 1$$

$$0 < V < 1$$

The marginal pdf of U is,

$$g(u) = \int_0^1 8uv^3 dv = 2u$$

The marginal pdf of U is,

$$g(v) = \int_0^1 8uv^3 du = 4v^3$$

since, $g(u,v) = g(u) \cdot g(v) = 8uv^3$

$\Rightarrow U$ and V are independent.

example: The pdf of the variable X and Y is, $f(x,y) = \frac{1}{2} x e^{-y}$, $0 < x < 2$, $y > 0$.

Find the marginal pdf of U .

Soln:

Let $u = x + y$

$v = x$

$y = u - v$

$\Rightarrow x = v$

Jacobian of transformation is,

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

$$|J| = 1$$

The required pdf is,

$$\begin{aligned} g(u,v) &= f(x,y) |J| \\ &= \frac{1}{2} x e^{-y} (1) \\ &= \frac{1}{2} v e^{-(u-v)} \end{aligned}$$

Domain,

$$0 < x < 2 \Rightarrow 0 < v < 2$$

$$y > 0 \Rightarrow u - v > 0 \Rightarrow u > v$$

$$\therefore g(u,v) = \frac{1}{2} v e^{-(u-v)} \quad ; 0 < v < 2, u > v$$

The marginal pdf of V is,

$$g(u) = \int \frac{1}{2} v e^{-(u-v)} dv \quad ; 0 < v < 2 \\ u > v \\ = \frac{1}{2} e^{-u} \int v e^v dv$$

when,

$$0 < v < 4$$

$$g(u) = \frac{1}{2} e^{-u} \int_0^u v e^v dv$$

$$= \frac{1}{2} e^{-u} (v e^v - e^v)_0^u$$

$$= \frac{1}{2} e^{-u} (u e^u - e^u + 1) \quad ; 0 < u < 2$$

when,

$$0 < v < 2$$

$$g(u) = \frac{1}{2} e^{-u} \int_0^2 v e^v dv$$

$$= \frac{1}{2} e^{-u} [v e^v - e^v]_0^2$$

$$= \frac{1}{2} e^{-u} (2e^2 - e^2 + 1)$$

$$= \frac{1}{2} e^{-u} (e^2 + 1) \quad 0 < u < \infty$$